

## ● APPLICANT DOSSIER &amp; PORTFOLIO

# FERREL RASHAD AKEYLA

## Software Engineer · Creative Technologist · Product Explorer

Information Systems graduate with a passion for creative web development, high-performance visual computing, and applied AI. Dedicated to transforming complex data into humane, accessible, and meaningful digital tools.

### 01 // HUMAN-CENTERED IMPACT

Solving community problems through real-time open data and accessibility.

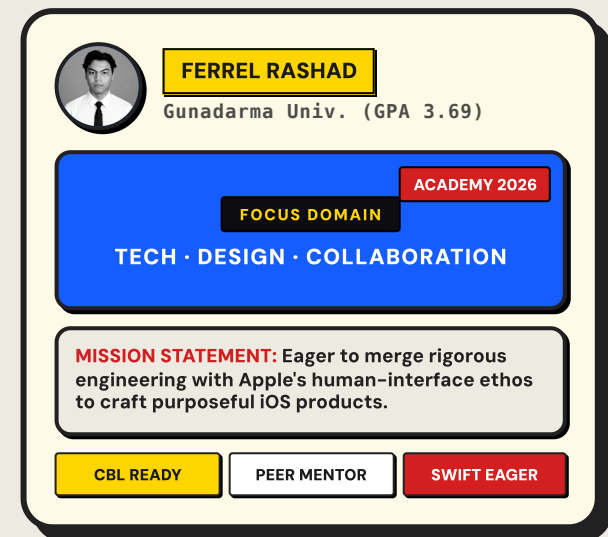
### 02 // TECHNICAL CRAFTSMANSHIP

60 FPS vector rendering, deep learning NLP, and interactive state management.

✉ ferrelrashadakeyla2014@gmail.com

🐙 github.com/FerrelHD

📍 Depok, West Java, Indonesia



## CHAPTER 01 // CAPABILITIES

# TECHNICAL CRAFTSMANSHIP & CREATIVE TOOLING

Disciplined foundations across system architecture, user interface design, and investigative research.

## </> UI & INTERACTIONS

- ★ React 19 & Next.js
- ★ TypeScript / Modern JS
- ★ GSAP & Framer Motion
- ★ Fluid Scroll & Micro-animations

→ Fluid, accessible, and responsive user interfaces.

## 🔧 ARCHITECTURE & DATA

- ★ Supabase & PostgreSQL
- ★ REST APIs & Live Telemetry
- ★ Laravel 11 Backend
- ★ Haversine & Geodesic Logic

→ Reliable pipelines connecting raw data to interface.

## 🛡️ APPLIED AI & RESEARCH

- ★ IndoBERT & Transformers
- ★ Sentiment NLP Classification
- ★ Python, Scikit-learn, PyTorch
- ★ Streamlit Analytics Dashboards

→ Extracting human insights and feedback patterns.

## 🗣️ PRODUCT & TEAMWORK

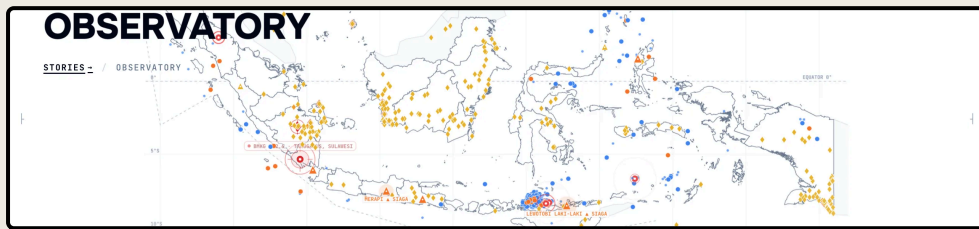
- ★ Figma Prototyping & Design
- ★ 3D Spatial Systems (Unity/Blender)
- ★ Peer Mentoring & Lab Tutoring
- ★ Cross-Functional Collaboration

→ Empathy for team dynamics and end-user clarity.

# FLAGSHIP CASE STUDIES: CHALLENGE-BASED SOLUTIONS

CBL STUDY  
CASES

Investigating real problems to deliver accessible, high-performance community tools.



SOCIAL IMPACT / OPEN DATA

React 19 • Canvas 2D • Telemetry

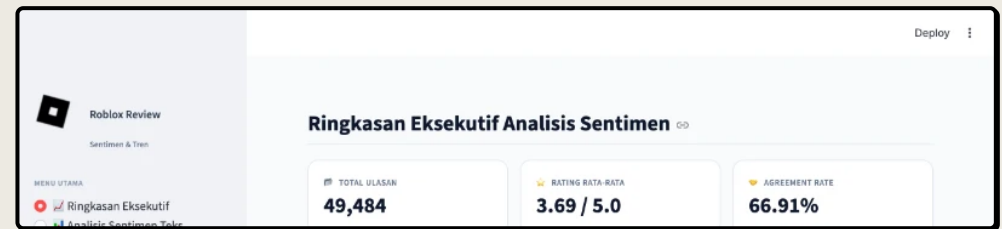
## NUSANTARA CRUSTAL OBSERVATORY & HAZARD TRACKER

**The Problem (Engage):** Indonesian citizens frequently receive fragmented, anxiety-inducing disaster alerts scattered across social media without contextual proximity.

**Investigation & Solution (Investigate & Act):** Unified live feeds from USGS, BMKG AutoGempa, NASA FIRMS, and PVMBG into a custom-built 60 FPS Canvas 2D vector map. Integrated Haversine geodesic calculation so users instantly know their danger clearance radius.

✓ 60 FPS Zero-Lag Mobile Visualization

Canvas 2D • TypeScript • Supabase



UNDERGRADUATE THESIS / NLP

IndoBERT vs SVM • Streamlit

## ANALISIS SENTIMEN ULASAN ROBLOX: SVM VS INDOBERT

**The Problem (Engage):** Sharp fluctuations and sudden waves of negative reviews hit the community between February and April 2026, leaving unclear root causes behind the vocal backlash.

**Investigation & Solution (Investigate & Act):** Fine-tuned IndoBERT (achieving 87.61% accuracy against informal slang) and built an exploratory dashboard cross-referencing complaint clusters with official developer changelogs to pinpoint the policy disruption.

✓ 87.61% Accuracy on Informal Language

PyTorch • IndoBERT • Streamlit

# INTERACTIVE SYSTEMS & DIGITAL EXPERIENCES

INTERACTION  
& 3D

Demonstrating passion for graphic fidelity, physics, and smooth client-side interaction.



INTERACTIVE WEB

GSAP / Physics

## MONACO GP TELE...

Fluid Ribbon Trail Bézier physics and dynamic telemetry HUD recreating real-world racing metrics.

Sub-step physics at 60–120 FPS



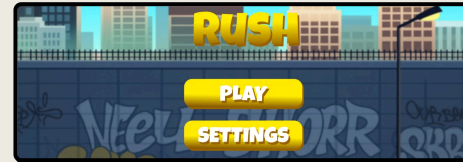
SIMULATION &amp; 3D

Roblox / Luau

## GUNUNG GEDE T...

Realistic topographic terrain modeling, dynamic weather systems, and stamina physics reaching 4M+ visits.

Team production & environment design



MOBILE PHYSICS

Unity / C#

## STREET RUSH (3D)

Fast-paced runner with custom collision matrices, procedural obstacle spawners, and mobile optimization.

Optimized mobile runtime



WEB PLATFORM

Laravel / Filament

## FERSYA PLATFORM

Streamlined administrative controls for small business owners with real-time stock sync and clean mobile layouts.

Usability for non-tech merchants

## LEARNING ASPIRATION

# READY TO LEARN, COLLABORATE, & CREATE

I view software development not merely as writing code, but as a medium to solve real challenges. My objective at the Apple Developer Academy is to absorb iOS design standards, collaborate across multidisciplinary teams, and build products that empower communities.

 **COLLABORATIVE GROWTH**

Experienced tutor and lab assistant eager to share technical insights and learn design thinking.

 **SWIFT & APPLE ECOSYSTEM**

Excited to translate Canvas 2D and reactive web architectures into native SwiftUI and CoreML.



[ferrelrashadakeyla2014@gmail.com](mailto:ferrelrashadakeyla2014@gmail.com)



[github.com/FerrelHD](https://github.com/FerrelHD)



[linkedin.com/in/ferrel-rashad-8a165514b/](https://linkedin.com/in/ferrel-rashad-8a165514b/)

 **ACADEMY ALIGNMENT**

PRIMARY ROLE IN ACADEMY

**Coder / Tech Explorer**

COMPLEMENTARY INTERESTS

**Human Interface Design & Product Strategy**

LEARNING OBJECTIVE

**Mastering SwiftUI, CoreML, and Apple Design Guidelines**

**READY FOR CHALLENGE-BASED LEARNING**